

SQL Exercise 1

2024/09/26

SQL Fundamentals: SELECT and filtering

① Selecting and filtering.

TABLE employees

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-20	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Makana	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

SQL

SELECT *

FROM employees;

② TABLE employees

Department

IT

HR

Finance

Marketing

SQL

SELECT DISTINCT department
FROM employees;

③ TABLE employees

first_name

Eva

Alice

Cathy

David

Brian

last_name

Naidoo

Green

Zulu

Makana

Lee

SELECT first_name,
last_name

FROM employees

ORDER BY salary DESC;

④ TABLE employees

id	first_name	last_name	salary
05	Eva	Naidoo	72 000
01	Alice	Green	70 000
03	Cathy	Zulu	65 000

```
SELECT id,
first_name,
last_name,
salary,
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

⑤ filtering WHERE

TABLE - employee

id	first_name	last_name	department
1	Alice	Greens	IT
5	Eva	Naidoo	IT

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';
```

⑥ TABLE employees

id	first_name	last_name	department	Salary
2	Brian	Lee	HR	45 000
4	David	Mordena	Marketing	50 000

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'salary' > 60000 AND 'Finance'
```

* please not no "ears" its just salary

① TABLE employees

id	first_name	last_name	department
2	Brian	Lee	HR
5	David	Makenna	Marketing

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = HR 'HR' OR department = 'Marketing';
```

② TABLE employees

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Makenna	Marketing

```
SELECT id, first_name, last_name, department
FROM employees
WHERE NOT department = 'IT';
```

③ TABLE employees

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Nadlov	IT

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

(a) TABLE employees

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70 000	Johannesburg
5	Evd	Naidoo	IT	72 000	Johannesburg

SELECT id, first_name, last_name, department, city, salary
FROM employees
WHERE department = 'IT' AND salary > 65000 AND
city = 'Johannesburg')